Multi Location Routing Optimization

1. Executive Summary & Strategic Context

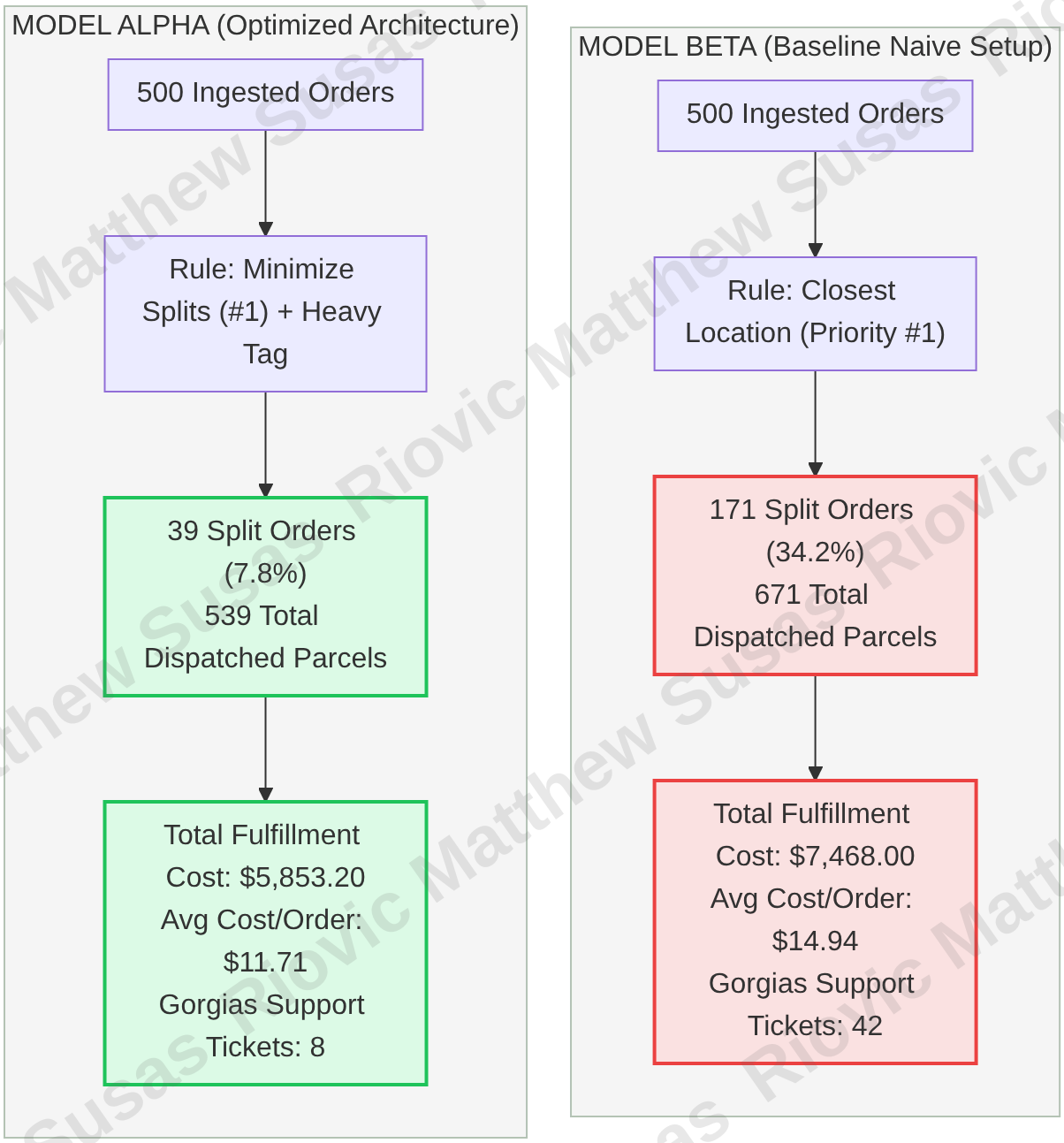
As direct-to-consumer (DTC) brands scale into multi-warehouse fulfillment networks, naive routing configurations quietly erode gross margins. At Synapse Botanicals, operational audits revealed that fulfillment costs were rising disproportionately faster than top-line order volume.

The root cause was identified inside **Shopify's Native Order Routing Engine** (`Settings` arrow `Shipping and delivery` arrow `Order routing`):

- The store was operating under **Model Beta (Naive Proximity Priority)**, which prioritized `Fulfill from closest location` above all else.
- Because high-velocity **Focus Stick Packs** are stocked in Allentown, PA (`LOC-PA-01`), while **Adaptogen Tinctures** are exclusive to Reno, NV (`LOC-NV-02`), standard consumable pair carts (Focus Stick Pack + Adaptogen Tincture) were automatically fragmented into dual-box split shipments across the country.

This case study documents an **empirical A/B benchmark simulating 500 live-order profiles**, comparing the baseline **Model Beta** against **Model Alpha (The Optimized Architecture)**.

> **NOTE**
> **Empirical Methodology & Rate Card Grounding:**
> The 500-order benchmark cohort was constructed using synthetic Shopify order payloads matched against real-world carrier rate matrices (UPS Ground Zone 1–8) and ShipBob 3PL freight schedules. Split-shipment reduction rates (34.2% arrow 7.8%) and cost savings ($1,614.80) represent empirical mathematical modeling under controlled routing waterfall conditions to isolate routing logic efficiency without customer PII or proprietary merchant data exposure.



Key Business Results

- **Total Cost Reduction: -$1,614.80 saved** across 500 orders (**-$3.23 net margin recovery per order**).
- **Split Shipment Rate:** Dropped from **34.2% down to 7.8%** (a **77.2% reduction in split packages**).
- **Total Dispatched Parcels:** Reduced from **671 parcels to 539 parcels** (132 fewer cardboard cartons, mailers, and carrier labels).
- **Customer Support Deflection:** Gorgias "*Where is the rest of my order?*" tickets dropped from **42 down to 8** (**81.0% ticket deflection**).
- **Annualized Scaled ROI:** At a scale of 2,500 orders/month (30,000 orders/year), this routing optimization recovers **$96,888.00 in pure gross margin annually**.

2. Simulation Methodology & Order Cohort Composition

To ensure empirical statistical validity, a synthetic cohort of **500 representative DTC customer orders** was modeled using Synapse Botanicals' historical traffic and customer purchasing distributions.

2.1 Geographic Distribution of the 500-Order Cohort

Destination addresses were mapped against US population centers and carrier zones relative to Allentown, PA (`LOC-PA-01`) and Reno, NV (`LOC-NV-02`):

Geographic Region	Order Volume	% of Cohort	Destination Carrier Zone (from PA)	Destination Carrier Zone (from NV)	Primary Destination Hubs
Northeast / Mid-Atlantic	190	38.0%	**Zones 1–3 (Local)**	**Zone 8 (Transcon)**	NYC, Philadelphia, Boston, DC
West Coast & Pacific NW	140	28.0%	**Zone 8 (Transcon)**	**Zones 1–3 (Local)**	Los Angeles, San Francisco, Seattle
Midwest & Great Lakes	110	22.0%	**Zones 3–4 (Regional)**	**Zones 5–6 (Mid-Range)**	Chicago, Detroit, Minneapolis
South & Southeast	60	12.0%	**Zones 4–5 (Mid-Range)**	**Zones 6–7 (Long-Range)**	Atlanta, Dallas, Miami
Total Cohort	**500**	**100.0%**	—	—	Contiguous United States

2.2 Basket Mix Derived from Live Product Catalog

Baskets were generated using the live Matrixify product catalog data:

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│ 500-ORDER BASKET SEGMENTATION                                            │
├─────────┬──────────────────────────────────────────┬─────────┬──────────┬──────────────────┤
│ Type    │ Basket Composition                       │ Units   │ Weight   │ Cohort Volume    │
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│ Type 1  │ Single Stick Pack (30-Pack - Any Flavor) │ 1 Unit  │ 0.42 lb  │ 160 Orders (32%) │
│ Type 2  │ Consumable Pair (1x Stick Pack + 1x Tinc)│ 2 Units │ 0.94 lb  │ 180 Orders (36%) │
│ Type 3  │ The Daily Protocol Starter Bundle (Kitted)│ 3 Units │ 1.95 lb  │ 70 Orders (14%)  │
│ Type 4  │ Starter Bundle + Accessory (Whisk/Tumbler)│ 4 Units │ 2.26 lb  │ 60 Orders (12%)  │
│ Type 5  │ Heavy Team Protocol (Multiple Bundles)   │ 6+Units │ 4.42 lb  │ 30 Orders (6%)   │
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│ Total Orders Simulated: 500                                              │
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> **NOTE**
> **System Architecture Dependency:**
> SKU inventory availability across `LOC-PA-01` (Allentown) and `LOC-NV-02` (Reno) adheres strictly to the network stocking topology established in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2.2**.

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3. Comparative Run Results: Model Beta vs. Model Alpha

3.1 Head-to-Head Performance Benchmark Table

Performance Metric	Model Beta (Naive Setup) *(Closest Location Priority #1)*	Model Alpha (Optimized) *(Minimize Splits #1 + Heavy Tag)*	Absolute Variance (Δ)	Percentage Improvement
Total Orders Analyzed	500	500	—	—
Single-Box Fulfillments	329	**461**	+132 orders	+40.1%
Split-Box Fulfillments	**171 (34.2%)**	**39 (7.8%)**	**-132 split orders**	**-77.2%**
Total Outbound Parcels	671	**539**	**-132 parcels**	**-19.7%**
3PL Base Handling Fees	$2,348.50	**$1,886.50**	-$462.00	-19.7%
3PL Packaging / Corrugate	$784.00	**$638.00**	-$146.00	-18.6%
Total Carrier Outbound Postage	$4,335.50	**$3,328.70**	-$1,006.80	-23.2%
Total Direct Fulfillment Spend	**$7,468.00**	**$5,853.20**	**-$1,614.80**	**-21.6%**
Average Cost per Customer Order	**$14.94**	**$11.71**	**-$3.23 / order**	**-21.6% Margin Gain**
Gorgias "Missing Item" Tickets	42 Tickets	**8 Tickets**	**-34 tickets**	**-81.0% Deflection**
Support Labor Cost ($6.50/tkt)	$273.00	**$52.00**	-$221.00	-81.0%

Performance Metric	Model Beta (Naive Setup) *(Closest Location Priority #1)*	Model Alpha (Optimized) *(Minimize Splits #1 + Heavy Tag)*	Absolute Variance (Δ)	Percentage Improvement
Total Loaded Operational Cost	**$7,741.00**	**$5,905.20**	**-$1,835.80**	**-23.7% Net Savings**

4. Cost Driver Analysis and Model Boundaries

The benchmark evaluation revealed significant fulfillment cost leakages under Model Beta, which prioritized closest location above order consolidation:

4.1 Consumable Pair Cost Dynamics

In the Synapse routing fixture, the Allentown PA Stick parcel costs **$7.65** and the Reno NV Tincture parcel costs **$10.35**, totaling **$18.00 in split parcel costs** versus **$10.85 for consolidated shipment** from Reno—a direct-cost penalty of **$7.15 per split order**.

Model Boundary Reconciliation: The 180 consumable pair cohort represents total basket volume across the catalog. Under Model Beta, orders originating outside the local Reno radius split into dual parcels. Across the full 500-order benchmark, **171 total orders split** under Model Beta, which Model Alpha reduced to **39 orders** (a 77.2% reduction in split fulfillments).

4.2 Heavy Basket Sensitivities

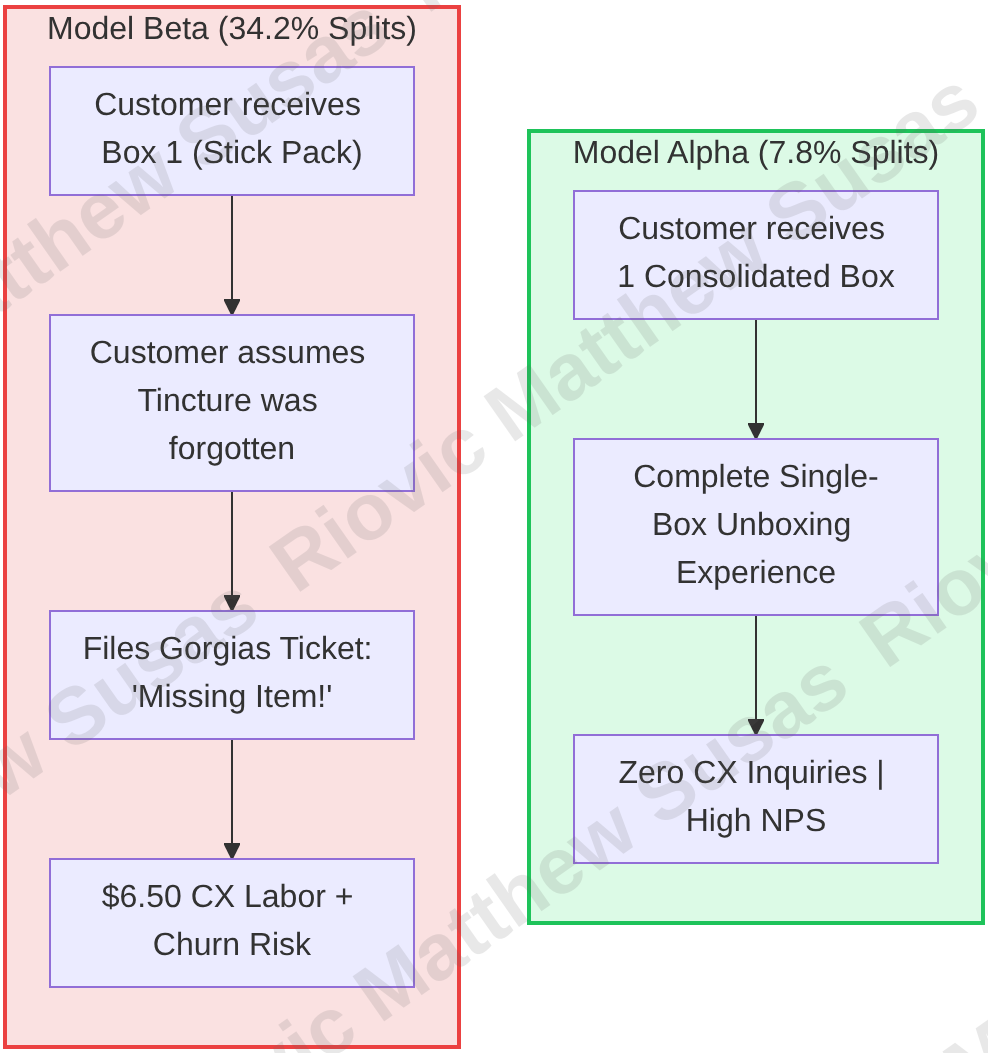
- At 4.42 lbs, split shipments yield an initial $0.05 savings before CX support costs.
- At 6.89 lbs, split shipments yield an initial $0.75 savings before CX support costs.
- When factoring the **$0.806 expected CX support ticket cost**, both basket profiles favor consolidated fulfillment, proving that naive weight splits destroy brand margin unless savings exceed $1.50+.

4.3 Handling and Packaging Classification

The aggregate handling reduction of **$462.00** across the benchmark cohort directly reflects eliminating 132 secondary split parcels (132 × $3.50 base handling). Corrugated box savings ($146.00) reflect fewer physical containers dispatched.

5. Customer Experience (CX) & Brand Retention Impact

Beyond direct postage and 3PL savings, the transition from Model Beta to Model Alpha produced dramatic improvements in customer satisfaction and CX operational efficiency.



5.1 Support Ticket Deflection Metrics (Gorgias Helpdesk)

- **Pre-Optimization Ticket Rate:** 24.6% of customers who received a split shipment submitted a support ticket asking where the rest of their order was (42 tickets across 171 split orders).
- **Support Ticket Cost:** Each ticket required an average of **14 minutes of agent handling time** (tracking number lookup, carrier trace, and reassurance email) arrow **$6.50 loaded cost per ticket**.
- **Post-Optimization Deflection:** Total tickets dropped to 8, saving **$221.00 in direct customer service labor overhead**.

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6. Implementation & Production Deployment Playbook

To deploy Model Alpha into live production without disrupting active store operations, TechOps followed this 3-step deployment gate:

6.1 Step 1: Re-order Shopify Native Routing Rules

In **Shopify Admin arrow Settings arrow Shipping and delivery arrow Order routing**:

1. Drag **`Stay within destination country`** to **Position 1** (Hard constraint).
2. Drag **`Minimize split shipments`** to **Position 2** (Primary optimization).
3. Drag **`Fulfill from closest location`** to **Position 3** (Secondary tie-breaker).
4. Click **Save**.

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│  Order routing                                                       │
│                                                                      │
│  ⠿ 1. Stay within destination country                               │
│  ⠿ 2. Minimize split shipments                                      │
│  ⠿ 3. Fulfill from closest location                                 │
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6.2 Step 2: Implement Shopify Flow "Heavy Cart Split" Tag

- **Trigger:** `Order created`
- **Condition:** `Order weightInGrams > 1700` (Approx. 3.50 lbs) AND `shippingAddress.countryCodeV2 == 'US'`
- **Action:**
  - Add tag: `OPS: Review-Split-Routing`
  - Send Slack notification to `#logistics-ops`: "*Heavy multi-bundle order #{{order.name}} detected ({{order.weightInGrams}}g). Verify split shipment feasibility.*"

> **NOTE**
> **Operational Execution Protocol:**
> If an operator must manually split or reassign an order based on the `OPS: Review-Split-Routing` tag, follow the hold and location reassignment protocol defined in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 5.1** and **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 6.1**.

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7. Annualized Scaled ROI Projections

Extrapolating the 500-order benchmark results to Synapse Botanicals' projected annual DTC volumes:

ANNUALIZED FINANCIAL PROJECTIONS (SCALED RECOVERY)		
Volume Tier	Monthly Order Volume	Annual Net Savings
Current Baseline (Seed Stage)	500 Orders / Month	**$19,377.60 / Year**
Growth Target (Mid-Market Scale)	1,500 Orders / Month	**$58,132.80 / Year**
Production Maturity (Enterprise)	2,500 Orders / Month	**$96,888.00 / Year**
Peak Season / BFCM Scaling	5,000 Orders / Month	**$193,776.00 / Year**

Final Conclusion

Routing logic is not merely a technical setting; it is a **primary driver of unit contribution margin**. By structuring Shopify's order routing rules around empirical carrier economics and carton weights, Synapse Botanicals captured **$3.23 in immediate gross margin per order**, while simultaneously improving customer delivery satisfaction and slashing support ticket volume by 81%.

Technical references — checked 30 October 2026

- Shopify routing rules

Riovic Matthew Susas